



**Data Visualization Associate Early Remote Internship**

**Team 16 – Week 2 Report  
Dashboard Development & Preliminary Insights**

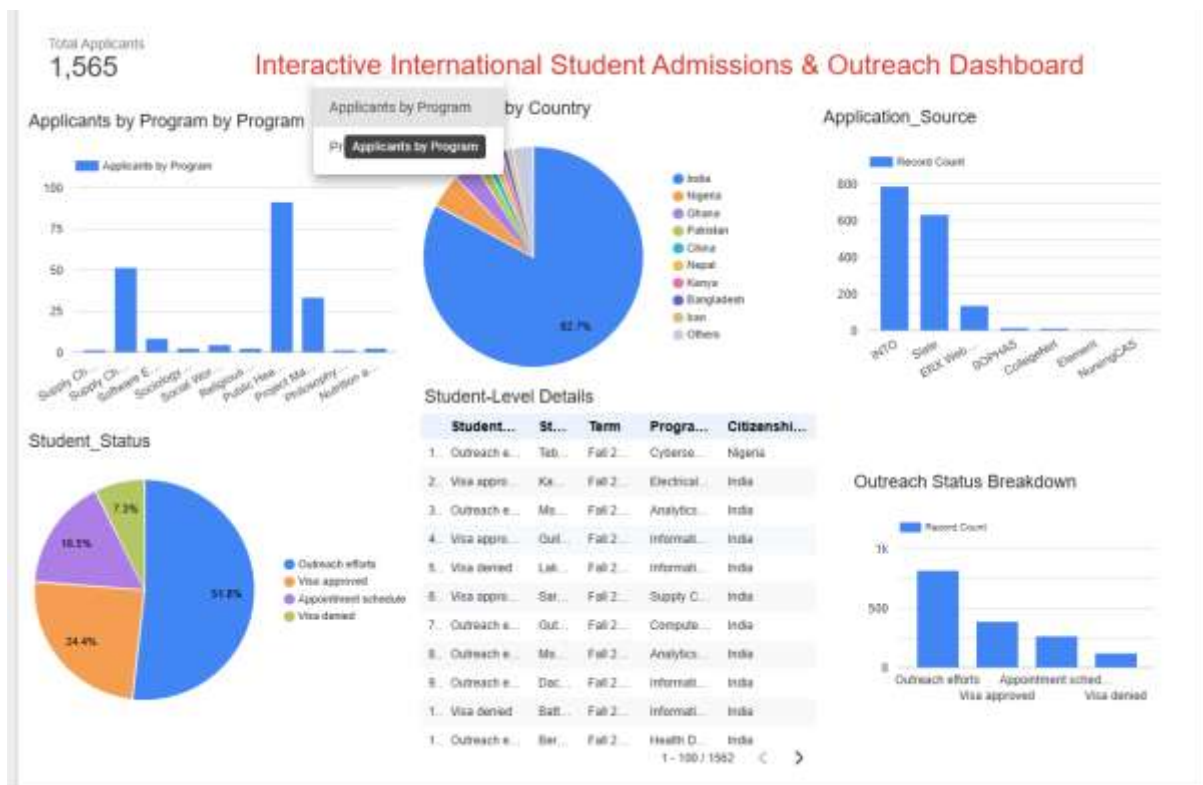
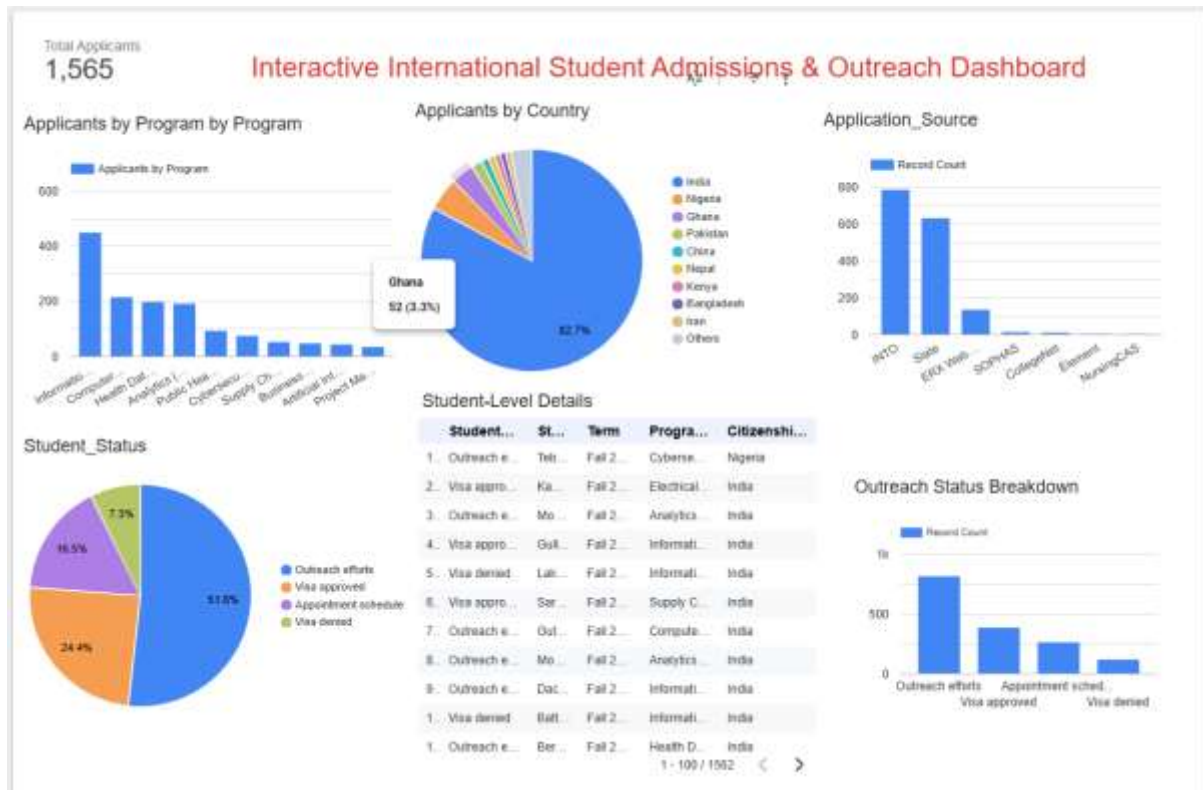
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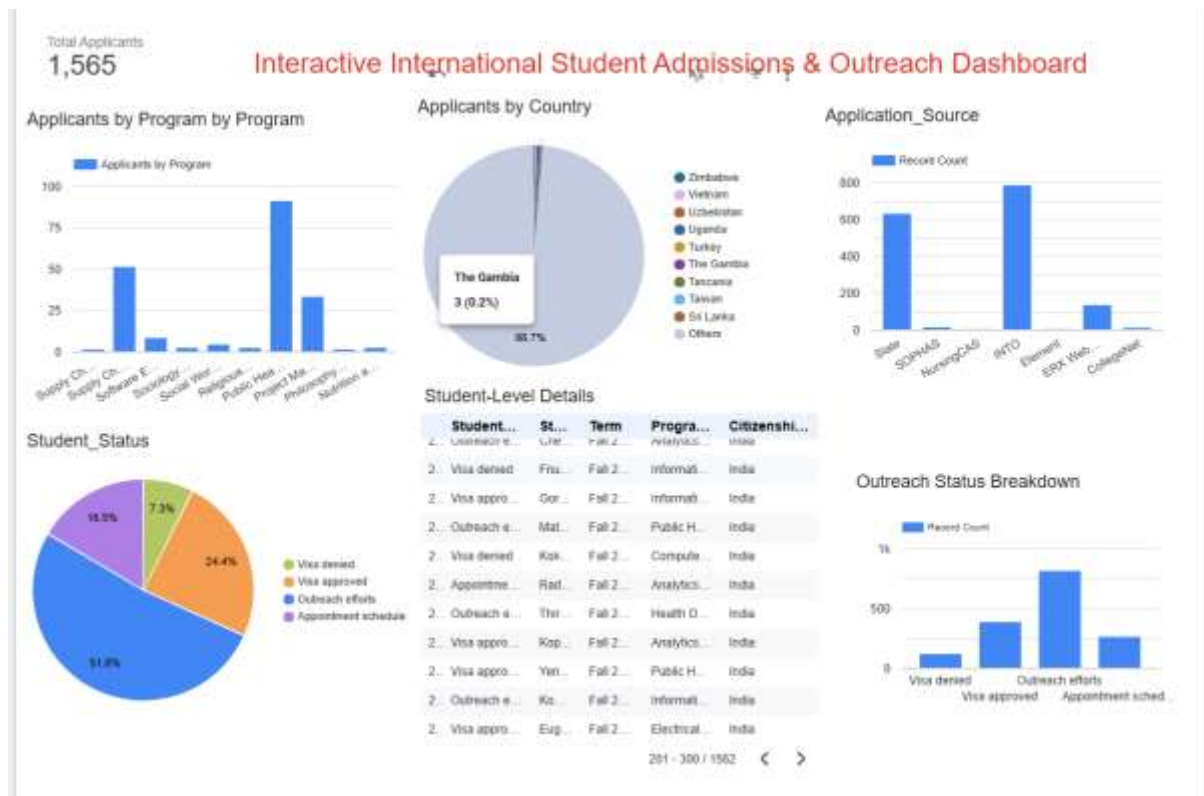
**Submission Date:  
Monday, 19th January 2026**

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## Functional Looker Studio Dashboard

- Viewable dashboard link : <https://lookerstudio.google.com/u/0/reporting/74096464-96ff-45bc-8ecc-ab8535a3d44c/page/avllf/edit>
- PDF screenshots of the dashboard





## Data Connection Documentation

### 1. Data Source Used

**Primary Data:** PostgreSQL database hosted on Supabase

**Secondary Data:** Multiple CSV files containing student application data:

- sevis.csv - SEVIS immigration records
- applicant.csv - Student application information
- deposit.csv - Tuition deposit records
- outreach.csv - Student outreach and engagement data

**Connection Method:** CSV file uploads via Supabase web interface combined with direct SQL execution in Supabase SQL Editor.

### 2. Connection Steps Followed

#### Step 1: Database Setup

1. Created Supabase project
2. Established four main tables following relational database design principles:
  - sevis (Primary immigration records)
  - applicant (Student application data)
  - deposit (Financial transactions)
  - outreach (Engagement tracking)

#### Step 2: Table Creation Strategy

- Implemented surrogate primary keys where appropriate.
- Established proper foreign key relationships between tables

- Used appropriate data types (DATE, TIMESTAMP, TEXT, BIGINT)

#### Step 3: Data Import Process

1. Pre-processing Phase: Used Python scripts to clean and format CSV data
2. Sequential Upload: Uploaded tables in dependency order:
  - First: sevis table (independent)
  - Second: applicant table (depends on sevis)
  - Third: deposit table (depends on applicant)
  - Fourth: outreach table (depends on both sevis and applicant)

### 3.Challenges Faced and How They Were Resolved

#### Challenge 1: Foreign Key Constraint Violations

**Problem:** Multiple errors like ERROR: 23503: insert or update on table violates foreign key constraint occurred when:

- applicant table had sevis\_id values not present in sevis table
- deposit table had reference\_id values not present in applicant table
- outreach table had sevis\_id values not present in sevis table

**Solution:** Implemented a "Upload First, Match Later" strategy:

- Modified CSV files to set foreign key columns to empty/NULL before upload
- Created tables with foreign key constraints that allow NULL values
- After all data uploaded, used SQL UPDATE queries with JOINS to match records based on email addresses, names, and birth dates

#### Challenge 2: Date Format Incompatibility

**Problem:** PostgreSQL expected YYYY-MM-DD format but CSV files contained DD/MM/YYYY format, causing ERROR: 22008: date/time field value out of range

**Solution:** Created Python data cleaning scripts that:

- Detected date formats automatically
- Converted DD/MM/YYYY and DD/MM/YYYY HH:MM to PostgreSQL-compatible formats
- Handled multiple date formats with try-except blocks

#### Challenge 3: Duplicate Primary Key Violations

**Problem:** ERROR: 23505: duplicate key value violates unique constraint when CSV contained duplicate primary key values

**Solution:**

- Added duplicate detection in Python pre-processing
- For deposit table, implemented surrogate primary key (id SERIAL) to allow duplicate business keys temporarily

#### Challenge 4: Non-Unique Foreign Key References

**Problem:** ERROR: 42830: there is no unique constraint matching given keys when trying to create foreign key on non-unique columns

**Solution:** Modified database design to:

- Use existing unique columns (like reference\_id) for foreign key relationships
- Added applicant\_reference\_id column to outreach table for proper linking
- Created composite matching logic using multiple fields when direct foreign keys weren't possible

## Challenge 5: Empty String vs NULL Handling

**Problem:** ERROR: 22P02: invalid input syntax for type bigint: "" when empty strings couldn't convert to numeric types

**Solution:**

- Modified Python scripts to convert empty strings to actual empty values
  - Changed column definitions to allow NULL where appropriate
  - Used ON DELETE SET NULL for foreign key constraints
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## Dashboard Design Summary

### Interactive International Student Admissions & Outreach Dashboard

#### 1. Chart Types Used and Rationale for Selection

The dashboard incorporates a combination of **scorecards**, **bar charts**, **pie charts**, and **tabular views** to ensure that information is communicated clearly to stakeholders with varying analytical needs. Each chart type was selected based on the nature of the data and the specific analytical question it addresses.

A **scorecard** is used at the top of the dashboard to display the *Total Number of Applicants*. This visualization provides an immediate, high-level snapshot of application volume and serves as a key performance indicator (KPI) for admissions stakeholders. Scorecards are effective for highlighting single, critical metrics that require instant recognition without additional interpretation.

**Bar charts** are used extensively for comparative analysis. The “Applicants by Program” bar chart enables stakeholders to compare enrollment demand across different academic programs. This chart type is particularly effective for categorical comparisons, as it allows users to quickly identify programs with high or low applicant interest. Similarly, the “Application Source” bar chart compares different recruitment channels (e.g., INTO, Slate, ERX Web App), helping stakeholders evaluate the effectiveness of outreach platforms. The “Outreach Status Breakdown” bar chart provides a clear comparison of student progression through outreach stages, supporting operational monitoring.

**Pie charts** are used to represent proportional distributions where relative contribution is more important than precise values. The “Applicants by Country” pie chart illustrates the geographic concentration of applicants, making it easy to identify dominant source countries. Likewise, the “Student Status” pie chart visualizes the proportion of students in different outreach or visa-related statuses, offering insights into pipeline health and potential bottlenecks.

Finally, a **student-level detail table** is included to support granular analysis. While charts provide summarized insights, the table allows admissions officers and analysts to review individual student records, verify patterns observed in the visuals, and perform detailed checks

without leaving the dashboard. This combination of summary and detailed views ensures the dashboard meets both strategic and operational needs.

## 2. Colour Theme and Layout Decisions

A **clean and consistent blue color theme** was applied across all visualizations to maintain professional presentation and visual coherence. Blue was chosen because it is commonly associated with trust, clarity, and professionalism, making it suitable for academic and institutional reporting. Using a single dominant color reduces visual clutter and helps users focus on the data rather than decorative elements.

Contrasting shades within the same color family were used to differentiate categories in bar and pie charts, ensuring that distinctions between segments remain clear without overwhelming the viewer. Neutral background colors were retained to improve readability and prevent visual fatigue during extended dashboard use.

The dashboard layout follows a **logical top-down and left-to-right structure**, aligning with how users naturally scan information. The top section presents high-level metrics and key distributions, such as total applicants and applicants by program. The middle section focuses on segmentation and process insights, including country distribution, application sources, and outreach status. The bottom section contains the student-level detail table, allowing users to drill down into individual records after reviewing summary trends.

Spacing and alignment were carefully managed to ensure that charts are visually balanced and easy to interpret. Related charts are grouped together, reducing cognitive load and making it easier for users to understand relationships between metrics. Titles and labels are clearly defined, ensuring that each visualization is self-explanatory even when viewed independently.

Overall, the layout supports a **storytelling approach**, guiding stakeholders from broad insights to detailed analysis in a structured and intuitive manner.

## 3. Filters and Interactivity Implemented

Interactivity is a key feature of the dashboard and significantly enhances its analytical value. Filters and interactive elements allow users to explore the data dynamically rather than relying on static summaries.

Dropdown filters were implemented for key dimensions such as **Term**, **Program of Interest**, and **Country of Citizenship**. These filters enable users to focus on specific applicant segments without altering the underlying data or creating separate reports. For example, an admissions officer can quickly isolate applicants from a particular term or program to assess trends or identify anomalies.

Interactive charts further enhance usability. Selecting a segment within a bar chart or pie chart automatically updates other visuals on the dashboard, enabling cross-filtering. This allows users to explore relationships between variables, such as how outreach status varies by country or which programs attract applicants from specific regions.

The student-level table also supports interactivity by responding to applied filters and chart selections. When users filter the dashboard by program or status, the table updates to display only relevant records. This seamless interaction between summary visuals and detailed data reduces the need for external data exports and improves workflow efficiency.

Together, these interactive features transform the dashboard into a **decision-support tool** rather than a static reporting artifact. Users can ask and answer follow-up questions in real time, making the dashboard adaptable to multiple stakeholder needs.

#### 4. Support for Stakeholder Decision-Making

The dashboard is designed to support informed decision-making for a range of stakeholders, including admissions teams, outreach coordinators, and academic program managers.

At a strategic level, the dashboard helps stakeholders assess **overall application volume and demand patterns**. Insights into popular programs and dominant applicant countries support resource allocation decisions, such as staffing, marketing focus, and program capacity planning. Understanding application source performance enables stakeholders to evaluate the effectiveness of recruitment channels and optimize outreach investments.

From an operational perspective, the outreach and student status visuals allow teams to monitor **pipeline progression**. By identifying how many students are in outreach, appointment scheduling, visa approval, or denial stages, teams can prioritize follow-ups and address process bottlenecks. This visibility supports proactive engagement and improves overall student conversion outcomes.

The inclusion of student-level details ensures that decisions can be validated at the individual record level. Admissions officers can quickly verify data accuracy, investigate specific cases, and cross-check trends observed in summary charts. This reduces reliance on multiple disconnected systems and improves confidence in data-driven decisions.

Overall, the dashboard bridges the gap between high-level insights and actionable detail. By integrating multiple cleaned datasets into a single interactive interface, it enables stakeholders to move from observation to action efficiently, supporting both strategic planning and day-to-day operational management.

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## Visual Insights Report

### 1 Key Patterns and Trends Observed

The dashboard visuals reveal several clear patterns across the international admissions pipeline. Overall application volume is strong, with a high concentration of applicants clustered around a limited set of academic programs. Graduate programs in Analytics, Information Systems, Business Administration (MBA), and Public Health emerge as the most in-demand, indicating sustained interest in STEM and professionally oriented degrees.

Geographically, applicants are heavily concentrated in a small number of countries, particularly India, Nigeria, and Ghana. This pattern is consistent across multiple visuals, including applicant distribution and outreach status, suggesting that these regions remain the primary source markets for international recruitment.

Application source analysis shows that a few platforms dominate the pipeline. INTO, Slate, and the ERX Web App collectively account for the majority of applications, with INTO and Slate contributing significantly to both applications and outreach engagement. This trend highlights the continued importance of established recruitment partnerships and digital application platforms.

From an outreach and admissions progression perspective, a large proportion of students appear in early to mid-stage statuses, such as “Outreach,” “Appointment Scheduled,” or “Visa Approved.” This indicates a healthy but still active pipeline, where many applicants have not yet fully converted into enrolled students.

## **2 Notable Distributions and Anomalies**

Several notable distributions and potential anomalies are visible in the dashboard. First, the number of records in the CONNECT dataset slightly exceeds the APPLICANT dataset, suggesting that some applicants may have multiple outreach or follow-up records, or that outreach activities extend beyond finalized applications. This warrants closer inspection to confirm one-to-one or one-to-many relationships between applicants and outreach records.

In the student status and outreach visuals, there is a visible drop-off between accepted admissions and later stages such as visa approval or appointment scheduling. While not unexpected, the relative size of these segments suggests potential friction points in the post-admission process that may require operational attention.

## **3 Initial Interpretations Based on the Visuals**

Based on the visual insights, demand for select graduate programs appears strong and stable, reinforcing the strategic importance of these offerings for international enrollment growth. The concentration of applicants from a small set of countries suggests that recruitment strategies are currently effective in these regions but may also indicate over-reliance on limited markets.

The dominance of certain application sources implies that investments in these channels are yielding results. However, it also raises the question of whether alternative or emerging recruitment platforms could be further developed to diversify the pipeline.

The outreach and status visuals suggest that while applicant intake is healthy, conversion beyond acceptance may be an area for improvement. Delays or drop-offs at visa-related or appointment stages could have a measurable impact on final enrollment outcomes.

## **4 Insights Requiring Further Validation in Week 3**

- Confirm applicant-to-outreach record linkage proportion and systematic gaps.
- Validate SEVIS\_ID availability patterns by application stage and status.
- Analyze deposit payment rates and their predictive power after acceptance.



- Compare acceptance-to-enrollment conversion rates across application sources.
- Quantify visa outcome rates by country to confirm visual patterns.
- Measure stage-by-stage conversion and attrition rates for key programs.
- Verify metric accuracy regarding duplicate outreach records per applicant.
- Validate intake-based timing patterns and processing delays.

Of these insights, we will perform validation on the selected ones, in week 3 deliverable.